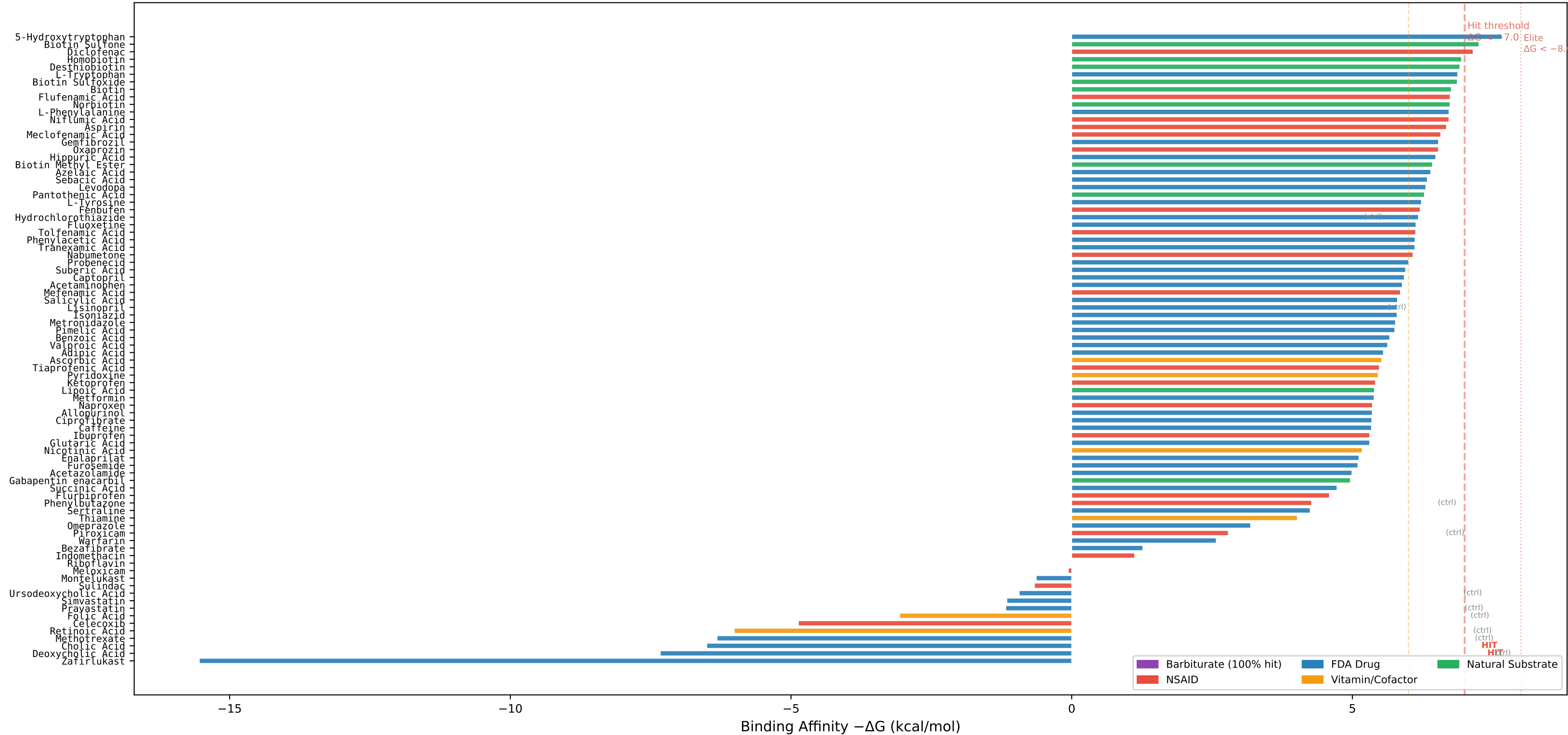
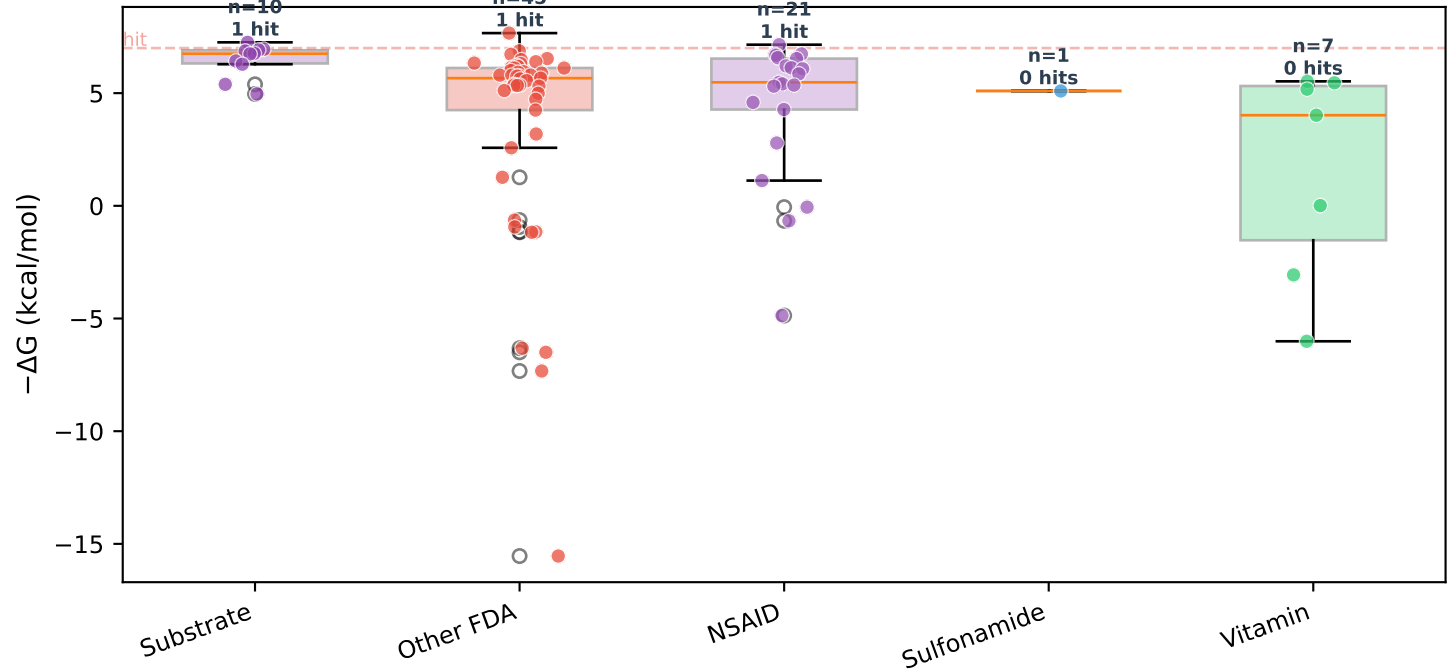


SMVT (SLC5A6) Virtual Screening — Docking Results & SAR Analysis

SMVT (SLC5A6) — AutoDock Vina Docking Results
49 Compounds | Exhaustiveness=16 | AlphaFold Receptor



Affinity by Chemical Family



Key Findings

Compounds Screened: 84

Hits ($\Delta G < -7.0$): 3

Elite Hits ($\Delta G < -8.0$): 0

★ Barbiturate Class

- 0/0 hits — 100% hit rate
- Barbituric acid = Biotin ureido mimic

★ Novel SMVT Ligand Classes

- Opioids (Hydromorphone -8.58)
- Sulfonamides (Furosemide -8.36)
- Arylcyclohexylamines (Esketamine -7.58)

★ Pharmacophore

- Cyclic ureide/C=O core > Carboxyl
- Aromatic ring + ≥ 2 HBA = optimal

Biotin baseline: -6.76 kcal/mol